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Effect of Robot Head Movement and its Timing on Human-Robot Interaction

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Abstract
Head movements can provide a significant amount of information in communication. This study investigated the social capabilities of a robot's head interaction, including the direction, movement, and timing of the movement. Using a newly designed robot, which has a minimal head movement mechanism in the three axes of x (pitch), y (roll), and z (yaw), we explored the participants' perception of the robot's head movements (i.e., nodding, shaking, and tilting) and movement timing (i.e., head movement prior to utterance and head movement simultaneous with utterance). The results revealed that head movements of the robot increased participants' perceptions of likeability, anthropomorphism, animacy, and perceived intelligence of the robot compared with the non-movement condition. When the robot performed head movement prior to utterance, the rating of perceived naturalness was higher compared with that when the robot's head movement occurred simultaneously with utterance. The findings imply that by implementing the features of head movements and movement timings, even simple robots that lack humanoid features can achieve better social interactions with humans.

Keywords Head movement · Nonverbal communication · Human-Robot interaction · HRI

1 Introduction

As intelligent social robots interact more closely with humans, their social abilities have become increasingly important. Designing robots that can communicate naturally with humans is one of the challenges in the field of Human-Robot Interaction (HRI), because communication with robots can be burdensome and unnatural [1]. One way to achieve natural interactions between humans and robots is to implement nonverbal communication. Humans in all cultures use nonverbal communication to provide interaction signals [2, 3] and focus on each other's heads to gather nonverbal cues. Nonverbal robot behaviors, including head movements, can improve collaboration between humans and robots [4, 5].

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Over the last few decades, to make HRI more natural, significant designs and studies have been conducted on humanoid robots that are capable of expressing human-like head movements [6, 7], and the social abilities of robots have been developed with advances in artificial intelligence [8]. Although the advances in artificial intelligence and engineering have made humanoid robots more human-like, they are not sufficiently sophisticated for natural HRI [9]. Moreover, several non-humanoid robots in our daily lives lack humanoid features. Owing to the limitations of embodiment, it is often impossible to exploit the straightforward nonverbal behaviors of humans in non-humanoid robots [10]. The behavior of non-humanoid robots should be easily interpretable in a social context [11]. Therefore, we should find modalities that can be implemented simply, not only for humanoid robots, but also for non-humanoid robots.

Despite the limitations of behavioral expressions, previous studies have revealed the potential of nonverbal expressions in non-humanoid robots [12, 13]. Zaga et al. [14] have shown that minimal gaze movements, which involve only the rotation of the head feature of the low-anthropomorphic robot, have a significant impact on social interaction. Other studies have also shown that the head features of social robots are beneficial for nonverbal expressions in life-like interfaces [15, 16]. In face-to-face communication, head poses and gestures provide communicative expressions (e.g., visual grounding, turn-taking, and yes/no answering) [17]. Furthermore, numerous experiments have shown that the head movements of robots have the potential to be more socially communicative [14, 18, 19] and enhance the perceived naturalness of robots [20]. However, few studies have investigated the timing of head movements, which are important for socially sophisticated expressions. Previous research has shown that separate timing of head and eye movements is significantly more natural than simultaneous movements [21]. Additionally, in the field of HRI, although real-time computation has been developed in many studies, the challenge of balancing accurate recognition and computation time delay still remains [22, 23]. Therefore, we focused not only on non-humanoid head movements but also on the decoupled timing of head movements and utterances of robots for social interaction with human users.

This study aims to assess the impact of a robot's minimal head movements on communicative expressions during HRI. Using a non-humanoid robot newly designed for this study, we designed experiments to simulate conversations between participants and the robot. Three types of head movements (nodding, shaking, and tilting) and the timing of head movement (i.e., movement prior to the utterance or movement simultaneous with the utterance) were evaluated.

The objectives of this study are as follows: (i) explore how head movements and movement timing of non-humanoid robots affect user perception, and (ii) address the challenge of smooth HRI through head movement and movement timing, which lack humanoid features. It was expected that robots with simple shapes would have better social interactions with human partners through the implementation of head movements.

2 Head Movement in Face-to-Face Interaction

Goffman defined face-to-face interaction as the reciprocal influence of individuals [24]. As a social value, people obtain impressions and information about individuals through their faces [25]. During face-to-face interactions, people can directly observe others, collect nonverbal cues [26], and use both verbal and nonverbal signals in the grounding process to understand others [27, 28]. Such signals are essential for the flow of meaning and emotional expression in face-to-face interactions [29]. During such interactions, people gather information through each other's signals, thereby facilitating better communication.

Previous research has shown that face-to-face interactions lead to more positive impressions of partners and greater self-other agreement than online interactions [30]. In addition, behavioral awareness in face-to-face environments enhances social interaction in group communication [31], thereby amplifying co-created meaning [32]. Through face-to-face interactions, people can observe nonverbal cues and understand each other better. Therefore, face-to-face interaction should be considered when designing social robots to facilitate better social interaction in HRI.

In face-to-face interaction, Nonverbal behaviors are self-presentational and convey a specific meaning to observers [24, 33]. They carry a significant portion of social meaning and emotional expression in face-to-face interaction [34, 35]. Through nonverbal behaviors, a large part of the initial assessment is observed visually [36], and social relationships, such as who we are and how we are related can be defined [37, 38]. Nonverbal communication can maintain communication fluency [39] and increase turn-transition speed [40]. Nonverbal communication skills are an essential part of social competence in our daily lives and help people understand social contexts [41].

In human conversations, head movements have predictable kinetic patterns and offer communicative functions [42, 43]. The head nodding movement, that is, moving the head vertically up and down, is used for affirmation in many cultures [42, 43]. Furthermore, the head shaking movement, that is, moving the head horizontally in side-to-side sweeps, is generally used to signal negation [42–44]. Additionally, the lateral head tilt is used to express uncertainty about an answer [42]. Head movement is a common and basic behavior in human communication that is observed by both speakers and listeners [45]. Head poses and gestures provide grounding cues during face-to-face communication [46]. The head movement is used as an element of turn-taking signals [47]. The listener's head nodding and direction play the role of a turn request [48, 49]. The listener's head movement is also observed in conversations as backchannel behavior [47, 49–51]. In addition, a speaker's head movements can capture the listener's attention [51]. Head movements have communicative patterns and functions in conversation.

2.1 Head Movement in Human-Robot Interaction

In HRI, the robot's head is an efficient tool for engaging in communication [15]. This is because people naturally see each other's faces during face-to-face communication. Recent study showed that a robot's nonverbal behavior such as head and body movements had a significant effect on trust perception in HRI [52]. In the interaction between humans and robots, the head movements of robots can enhance the user's perceptions of anthropomorphism [53–55], animacy [14], likeability [14, 53, 54, 56], and naturalness [57] of robots. Anthropomorphism [58] and animacy [59] are significant factors for enhancing HRI. In addition, animacy is significantly correlated with the perceived intelligence of the robots [59]. Although there have been many studies on the impact of robot head movements on user perception, few studies have investigated the independent head movements that can be implemented not only in humanoid robots but also in non-humanoid robots due to simple form and movement.

In this study, we explored the effects of a robot's head movements on the user perceptions of anthropomorphism, animacy, likeability, and intelligence by comparing them with three head movement conditions: nodding (A), shaking (B), and tilting (C). We hypothesized that minimal head movements of non-humanoid robots would positively affect user perceptions of anthropomorphism, animacy, likeability, and intelligence of the robot.

H1. The minimal head movement of the non-humanoid robot will increase the participants' perceptions of anthropomorphism (H1a), animacy (H1b), likeability (H1c), and intelligence (H1d) of the robot.

2.2 Timing of Robot Head Movement in Turn-taking Gap

Turn-taking is a fundamental speech-exchange system [60]. It is not only a formal procedure but also has social meaning and relationship [49, 61]. Both the speaker and listener focus on the partner's signals and engage in turn management [62]. In human-human conversation, the turn-taking transition, that is, speech exchange from one speaker to the next, is commonly tightly synchronized [63]. Turn-taking gaps in human conversations average approximately 200 ms [40, 64], and they are not recognized as silent gaps [65]. However, long response delays of the order of 2 s and uncomfortable silent gaps during turn-taking are seen in human-robot conversations [66]. Turn-taking has social meaning, and natural turn-taking is not perceived as an uncomfortable gap. Natural turn-taking between humans and robots should be considered while designing social robots.

According to Hadar et al. [67], in human conversations, head-posture shifts usually begin prior to speech onset and indicate the commencement of speech. Through head movements prior to speech, listeners can claim their speaking turns and prepare to speak [47, 67, 68]. In addition, head movements tend to follow hesitation pauses and occur before the commencement of speech [69]. In summary, head movement prior to speech plays the role of turn requesting, thus indicating speech preparation, and head movement during speech gaps, such as vocal hesitation, is common in human conversations.

In a human-robot conversation, the robot's head movements prior to utterance may play roles similar to those in a human-human conversation. We hypothesized that head movements prior to utterances (Fig. 1) may reduce the long response delay problem and show a higher perceived naturalness than head movements simultaneous with utterances.

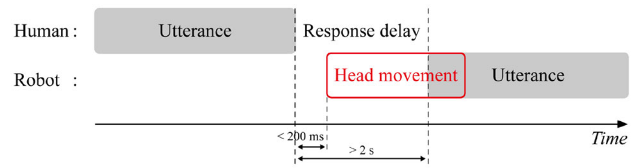

Fig. 1 Example of head movement prior to utterance

H2. Head movements prior to utterances will show a higher rating of perceived naturalness than head movements simultaneous with utterances.

Previous research has shown that the likeability of robots improves with the naturalness of their speech [70]. Another study suggested a strong correlation between perceived intelligence and naturalness [71]. We expected the perceived naturalness of head movements to correlate positively with likeability and perceived intelligence.

H2-1. The perceived naturalness of a robot's head movement may be positively correlated with its likeability (H2-1a) and perceived intelligence (H2-1b).

3 Method

3.1 Participants

Ninety participants (57 male, 33 female) with a mean age of 31.39 years (SD = 3.48, range 24-38) were recruited. None of the participants were visually or hearing-impaired. On average, the participants spent a total of 12.4 min (SD = 1.66, range 9-15), 3.5 min (SD =.50, range 3-4) on the actual task, and 8.9 min (SD = 1.65, range 5-11) on the questionnaire. This study was approved by the Institutional Review Board of Seoul National University.

3.2 Design

We investigated how a robot's head movements affect the participants' perception of the robot in a human-robot conversation. There were three movement conditions (nodding, shaking, and tilting) and two timing conditions (head movement prior to utterance and head movement simultaneous with utterance). Each participant was randomly assigned to one of three head-movement conditions: nodding (A), shaking (B), and tilting (C). The participants were divided into groups of 30 people for each movement condition. For each movement condition, conversation scripts were prepared according to the specific meaning of the robot's head movement such as affirmation of head nodding, negation of head shaking, and uncertainty of head tilting.

Each movement condition included two timing variables: head movement prior to the utterance and head movement

Table 1 Experimental Design

Robot's response	*N*	Conditions Movement	*Timing*	*N*
A (Affirmation)	*30*	Non-movement *(Control)*		
		Head nodding *(Treatment)*	AP *(Prior)*	15
			AS *(Simultaneous)*	15
B (Negation)	*30*	Non-movement *(Control)*		
		Head shaking *(Treatment)*	BP *(Prior)*	15
			BS *(Simultaneous)*	15
C (Uncertainty)	*30*	Non-movement *(Control)*		
		Head tilting *(Treatment)*	CP *(Prior)*	15
			CS *(Simultaneous)*	15

simultaneous with the utterance. In the condition involving head movement prior to the utterance, the robot at its turn, performed head movement within 200 ms and utterance within 2 s after the participant finished speaking. In the condition involving head movement simultaneous with the utterance, the robot at its turn, simultaneously performed both head movements and utterances within 2 s after the participant finished speaking. Participants grouped as per each movement condition were further divided into 2 groups of 15 participants each for every timing condition. The data on the number of participants for each condition is listed in Table 1.

3.3 Procedure

The participants, upon signing the consent forms were informed of the experimental procedure and provided with an example of the dialogue script they could use when talking to the robot. As given in the dialogue script, for the head nodding movement (A), when a participant said to the robot, "Hey robot, I would like to set an alarm," the robot responded affirmatively, "Ok, at what time would you like to set the alarm?". In case of the shaking movement (B), when a participant said to the robot, "Hey robot, tell me my account password," the robot responded in negation, "Sorry, this is not possible. Do you require anything else?" For the tilting movement (C), when a participant said to the robot, "Hey robot, I need to book a train ticket to Seoul," the robot responded with a doubt, "I didn't hear you. Can you rephrase your question?"

Participants were randomly assigned to one of the three movement conditions and asked to converse with the robot on the screen (Fig. 2). They spoke to the robot according to the script and received responses from it. In total, two rounds of conversations were conducted. In the first round of conversation, the robot responded only with an utterance (non-movement condition), whereas in the second, it responded with both head movement and utterance (head movement condition). At the end of each round of conversation, the participants were asked to complete a questionnaire.

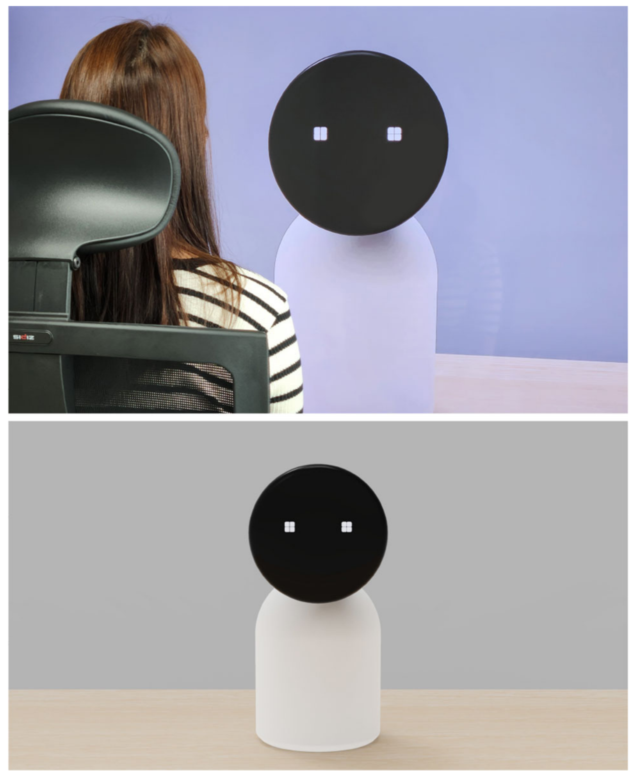

Fig. 2 Experiment with conversation simulation using 75-inch monitor

3.4 Materials

To specifically focus on the head movement in the experiment, we designed a non-humanoid robot with a simple-shaped body and minimal head movement mechanism along the three axes: x (pitch), y (roll), and z (yaw). Figure 3 shows the robot design. This mechanism enables the robot to perform three types of head movements (Fig. 4). The robot has the following dimensions: height of 445 mm, width of

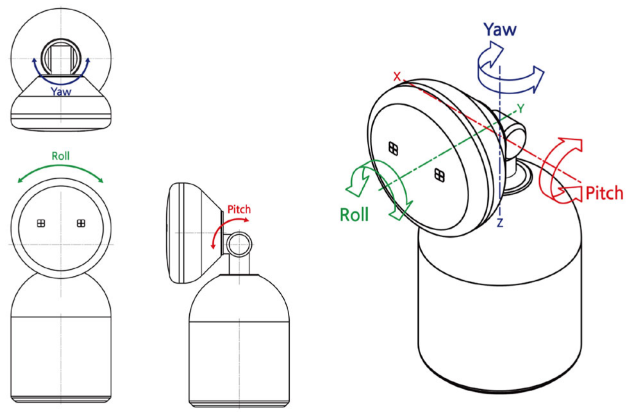

Fig. 3 Isometric drawing of the robot design and mechanism

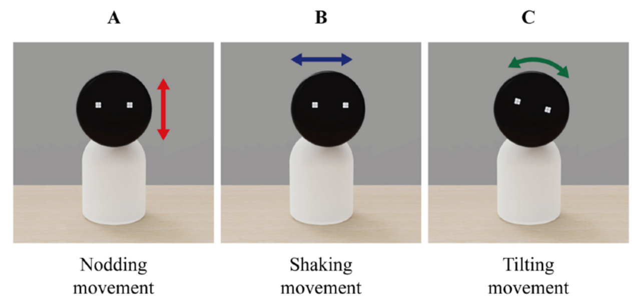

Fig. 4 Three types of head movements in the experiment

200 mm, depth of 245 mm, and a circular head with a diameter of 200 mm.

During the experiment, the robot performed three types of head movements according to predefined scenarios. In this experiment, we used a virtual environment to effectively compare the timing variables of the robot's head movements. Each type of movement was actuated with 3D computer graphic simulation. The head nod movement was performed for a total of 2.8 s by sequentially rotating -10 degrees, 10 degrees, -10 degrees, and 10 degrees for 0.7 s each based on the x-axis. The head shake movement was performed for a total of 2.8 s by sequentially rotating 15 degrees, -20 degrees, 15 degrees, and -10 degrees for 0.7 s each based on the z-axis. The head tilting movement was performed for a total of 1.4 s by sequentially rotating 10 degrees and -10 degrees for 0.7 s each based on the y-axis. Each animation was rendered at 30 frames per second (fps) using KeyShot, a 3D rendering software. We then added the robot's voice, created with text-to-speech (TTS), and matched the audio and video sources using Adobe Premiere Pro software. Each video source was created by configuring the robot's movement (i.e., nodding, shaking, or tilting), utterance (i.e., affirmation, negation, or uncertainty), and timing gap (i.e., no delay or 2 s). In the head movements prior to utterance condition, there was no delay before the head movement. The simultaneous timing condition included 2 s delay preceding head movement and utterance in the video source. The experimenter manually played the video source by pressing the spacebar on the paused screen immediately after the participant finished speaking one of the provided dialogue examples. After several pilot tests, all types of robot response conditions (3 (movements) × 2 (timings)) were prepared.

3.5 Measures

Anthropomorphism. Anthropomorphism refers to the human characteristics of non-human things [72]. It is a significant factor in predicting trust in robots [58] and influences users' intentions to use robots [73]. We used the Godspeed questionnaire [72] to measure the participants' perceptions of the anthropomorphism of the robot.

Animacy. Animacy refers to how users perceive animated, life-like movement [72]. We used the Godspeed questionnaire [72] to measure participants' perceptions of the robot's animacy.

Likeability. Likeability measures the degree of positive impressions of robots. We used the Godspeed questionnaire [72] to measure the participants' perception of the likeability of the robot.

Perceived intelligence. This concept refers to how much the robots were perceived as smart and intelligent. The Godspeed questionnaire [72] was used to measure the perceived intelligence of the robot.

Perceived naturalness. To investigate how the participants perceived naturalness according to the robot's head movement, the degree of perceived naturalness was measured on a 5-point Likert scale based on a previous study [20].

4 Results

4.1 Manipulation Check

Manipulation checks for each type of head movement (treatment) were analyzed by using a one-sample t-test. We measured the participants' perception rating of each type of head movements on a 5-point Likert scale: 5 = "strong" to 1 = "weak." The results revealed that the robot had significantly different head-movement conditions: nodding ($T(29) = 26.49$, $p < .001$), shaking ($T(29) = 26.49$, $p < .001$), and tilting ($T(29) = 29.57$, $p < .001$). The mean and standard deviation values of the results were: nodding ($M = 4.83$, $SD = .38$), shaking ($M = 4.83$, $SD = .38$), and tilting ($M = 4.87$, $SD = .35$).

Table 2 Internal consistency of measures in the experiment

Measure	Number of items	Cronbach's α
Anthropomorphism	5	.97
Animacy	6	.98
Likeability	5	.95
Perceived intelligence	5	.95

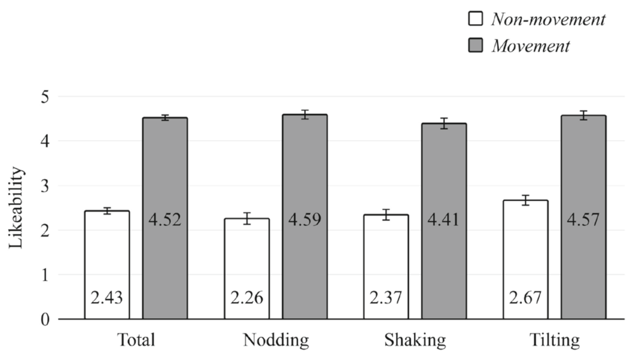

Fig. 5 Mean likeability scores in the experiment

4.2 Reliability

Cronbach's alpha was calculated to check the internal consistency of the participants' responses to anthropomorphism, animacy, likeability, and perceived intelligence. The results showed that all the measures were acceptable. Cronbach's alpha values for each measure in the experiment are presented in Table 2.

4.3 Results of H1

We investigated the effect of the head movements of the robot on the participants' perception of the robot. A within-subjects t-test was used to examine each type of head movement independently: non-movement (control) vs. head movement (treatment). To summarize overall, all the head movements, including nodding (A), shaking (B), and tilting (C), had a statistically significant effect on likeability ($t(89) = 21.92$, $p < .001$, $d = .90$). Results revealed that when the robot performed head movements during conversation, likeability was higher than that in the non-movement condition. Figure 5 indicates that likeability increased for all types of head movements: nodding, shaking, and tilting. The robot's head movements led to a positive impression of the robot during HRI.

For all types of head movement, the anthropomorphism ($t(89) = 47.22$, $p < .001$, $d = .64$), animacy ($t(89) = 57.81$, $p < .001$, $d = .54$), and perceived intelligence ($t(89) = 21.78$, $p < .001$, $d = .88$) of the robot showed significant differences between the head movement and non-movement conditions. The total head movement (M = 4.50, SD =.44) had a higher anthropomorphism rating than the non-movement (M = 1.34, SD =.42) of the robot. For animacy, total head movement (M = 4.69, SD =.37) was rated higher than non-movement (M = 1.43, SD =.40). Total head movement (M = 4.52, SD =.60) had a higher rating of likeability than non-movement (M = 2.43, SD =.66). For perceived intelligence, total head movement (M = 4.40, SD =.61) had a higher rating than non-movement (M = 2.39, SD =.60). The results revealed that all types of head movements of the robot significantly increased the participants' perception of anthropomorphism, animacy, likeability, and perceived intelligence. The mean and standard deviation values of the experimental variables are listed in Table 3.

4.3.1 Head Nodding

A within-subjects t-test revealed that the head nodding movement had statistically significant effects on the participants' perceptions of anthropomorphism ($t(29) = 23.75$, $p < .001$, $d = .72$), animacy ($t(29) = 30.76$, $p < .001$, $d = .59$), likeability ($t(29) = 12.96$, $p < .001$, $d = .98$) and intelligence ($t(29) = 11.89$, $p < .001$, $d = 1.01$) of the robot.

The head-nodding movement (M = 4.47, SD =.46) had a higher anthropomorphism rating than the non-movement (M = 1.33, SD =.52) of the robot. For animacy, the head-nodding movement (M = 4.69, SD =.41) was rated higher than non-movement (M = 1.40, SD =.46). The head-nodding movement (M = 4.59, SD =.56) had a higher rating of likeability than non-movement (M = 2.26, SD =.71). For perceived intelligence, the head-nodding movement (M = 4.45, SD =.59) had a higher rating than non-movement (M = 2.28, SD =.69). The results indicated that the head nodding movement significantly increased participants' perceptions of anthropomorphism, animacy, likeability, and perceived intelligence.

4.3.2 Head Shaking

A within-subjects t-test revealed that the head shaking movement had statistically significant effects on the participants' perceptions of anthropomorphism ($t(29) = 29.68$, $p < .001$, $d = .60$), animacy ($t(29) = 41.63$, $p < .001$, $d = .44$), likeability ($t(29) = 12.89$, $p < .001$, $d = .87$) and intelligence ($t(29) = 14.74$, $p < .001$, $d = .76$) of the robot.

The head-shaking movement (M = 4.48, SD =.40) had a higher anthropomorphism rating than the non-movement (M = 1.25, SD =.29) of the robot. For animacy, head-shaking movement (M = 4.73, SD =.31) was rated higher than non-movement (M = 1.37, SD =.29). The head-shaking movement (M = 4.41, SD =.67) had a higher rating of likeability than non-movement (M = 2.37, SD =.63). For perceived intelligence, the head-shaking movement (M = 4.29, SD =.69) had a higher rating than non-movement (M = 2.25,

Table 3 Mean and standard deviation values of experimental variables

Robot's response	Movement	Anthropomorphism *Mean (SD)*	Animacy *Mean (SD)*	Likeability *Mean (SD)*	Perceived intelligence *Mean (SD)*
Affirmation	Head nodding	4.47 (0.46)	4.69 (0.41)	4.59 (0.56)	4.45 (0.59)
	Non-movement	1.33 (0.52)	1.40 (0.46)	2.26 (0.71)	2.28 (0.69)
Negation	Head shaking	4.48 (0.40)	4.73 (0.31)	4.41 (0.67)	4.29 (0.69)
	Non-movement	1.25 (0.29)	1.37 (0.29)	2.37 (0.63)	2.25 (0.49)
Uncertainty	Head tilting	4.56 (0.46)	4.64 (0.38)	4.57 (0.57)	4.47 (0.55)
	Non-movement	1.43 (0.40)	1.51 (0.44)	2.67 (0.58)	2.63 (0.55)
Total	Head movement	4.50 (0.44)	4.69 (0.37)	4.52 (0.60)	4.40 (0.61)
	Non-movement	1.34 (0.42)	1.43 (0.40)	2.43 (0.66)	2.39 (0.60)

SD =.49). The results showed that the head-shaking movement significantly increased the participants' perceptions of anthropomorphism, animacy, likeability, and perceived intelligence.

4.3.3 Head Tilting

A within-subjects t-test revealed that the head tilting movement had statistically significant effects on the participants' perceptions of anthropomorphism ($t(29) = 28.73$, $p < .001$, $d = .60$), animacy ($t(29) = 30.66$, $p < .001$, $d = .56$), likeability ($t(29) = 12.48$, $p < .001$, $d = .84$) and intelligence ($t(29) = 11.76$, $p < .001$, $d = .85$) of the robot.

The head tilting movement ($M = 4.56$, $SD = .46$) had a higher anthropomorphism rating than the non-movement ($M = 1.43$, $SD = .40$) of the robot. For animacy, the head-tilting movement ($M = 4.64$, $SD = .38$) was rated higher than non-movement ($M = 1.51$, $SD = .44$). The head-tilting movement ($M = 4.57$, $SD = .57$) had a higher rating of likeability than non-movement ($M = 2.67$, $SD = .58$). For perceived intelligence, the head-tilting movement ($M = 4.47$, $SD = .55$) had a higher rating than non-movement ($M = 2.63$, $SD = .55$). The results revealed that the head-tilting movement significantly increased the participants' perceptions of anthropomorphism, animacy, likeability, and perceived intelligence.

4.4 Results of H2 to H2-1

4.4.1 Timing of Head Movement

A between-subjects t-test was conducted to examine the effect of the timing adjustment of the head movement: head movement prior to utterances versus head movement simultaneous with utterances. The timing adjustment of total head movements, including nodding, shaking, and tilting, showed a significant effect on perceived naturalness ($t(69.72) = 4.55$, $p < .001$, $d = .58$), wherein head movement prior to utterance ($M = 4.80$, $SD = .40$) was rated higher than head movement simultaneous with utterance ($M = 4.24$, $SD = .71$).

The head-shaking movement prior to the utterance was statistically significant ($t(23.21) = 4.89$, $p < .001$, $d = .56$). Although the head nodding ($t(28) = 1.56$, $p = 0.13$, $d = .47$) and tilting ($t(28) = 1.83$, $p = .08$, $d = .60$) movements prior to the utterance were not statistically significant, they positively affected the perceived naturalness. The head-nodding movements prior to utterance ($M = 4.80$, $SD = .41$) had higher perceived naturalness than the head-nodding movements simultaneous with utterance ($M = 4.53$, $SD = .52$).

The head shaking movement prior to utterance ($M = 4.80$, $SD = .41$) had a higher perceived naturalness than the head-shaking movement simultaneous with utterance ($M = 3.80$, $SD = .68$). Head-tilting movement prior to the utterance ($M = 4.80$, $SD = .41$) had a higher perceived naturalness than head-tilting movement simultaneous with the utterance ($M = 4.40$, $SD = .74$) (Table 4).

4.4.2 Correlations with Perceived Naturalness

To examine how perceived naturalness is related to likeability and perceived intelligence of a robot, we measured the correlations between them. Table 5 presents the significantly positive correlations between perceived naturalness and other variables such as likeability and perceived intelligence.

Table 4 Mean and standard deviation values of timing variables in the experiment

Head movement	Prior movement of utterance *Mean (SD)*	Simultaneous movement with utterance *Mean (SD)*
Head nodding	4.80 (.41)	4.53 (.52)
Head shaking	4.80 (.41)	3.80 (.68)
Head tilting	4.80 (.41)	4.40 (.74)
Total	4.80 (.40)	4.24 (.71)

Table 5 Correlations between perceived naturalness and other variables

Variable	*Perceived naturalness*	Likeability	*Perceived intelligence*
Perceived naturalness	1.00		
Likeability	.47**	1.00	
Perceived intelligence	.53**	.78**	1.00

**$p < .01$

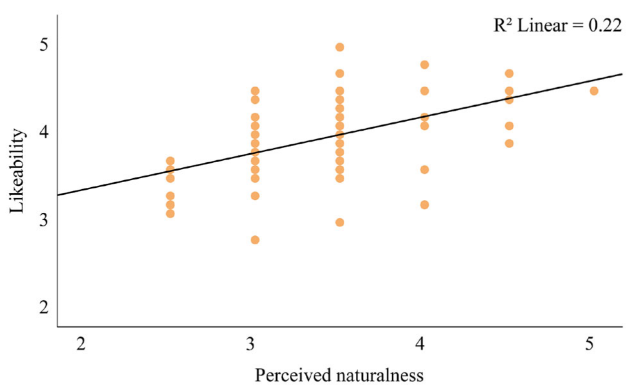

Fig. 6 Correlation between perceived naturalness and likeability

A positive correlation ($R^2 = 0.22$, $p < .01$) was observed between the perceived naturalness and likeability of the robot (Fig. 6) and also ($R^2 = 0.28$, $p < .01$) between the perceived naturalness and perceived intelligence of the robot (Fig. 7).

5 Discussion

In this study, we focused on the impact of the robot's head movements and movement timing. We used a newly designed simple robot for this experiment, focusing on the robot's independent head movements without other nonverbal behaviors such as facial expressions and body posture. Independent head movements are very important in HRI because many non-humanoid robots have limitations in rich physical expression. It can be applied to not only humanoid robots but also various types of non-humanoid robots. Although humanoid robots are becoming more and more similar to humans, there are still many burdens and limitations when using them in daily life. Therefore, a simple non-verbal expression such as independent head movement is important in HRI.

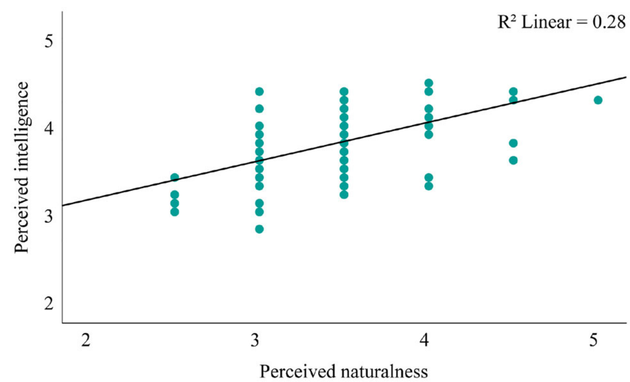

Fig. 7 Correlation between perceived naturalness and perceived intelligence

Furthermore, this study explored effects of the separate timing of a robot's head movements and utterances on users' perception of naturalness. This can reduce uncomfortable moments when users wait for robot response delays. Previous research in the field of HRI has developed real-time computation, but there are still time delays especially in complex and dynamic environments. Therefore, it is beneficial for the robot to perform head movements before utterance to fill the response gap. Because humans typically use head movements when hesitating to say the next word, a robot's head movement prior to utterance can be perceived as a natural conversational movement. We therefore investigated the decoupled timing of the robot's head movements and utterances.

The findings of this study are as follows: (i) all types of the robot's head movements (i.e., nodding, shaking, and tilting) significantly affected user perception in HRI, and (ii) the timing of the head movement prior to utterance had a significant effect on perceived naturalness in human-robot conversations.

First, Participants experienced more anthropomorphism (H1a), animacy (H1b), likeability (H1c), and perceived intelligence (H1d) of the robot with the head movement of the robot than the non-movement (utterance only) condition. In terms of anthropomorphism (H1a) and animacy (H1b), we observed that all types of head movements, nodding (A), shaking (B), and tilting (C), had positive effects on anthropomorphism ($t(89) = 47.22$, $p < .001$, $d = .64$) and animacy ($t(89) = 57.81$, $p < .001$, $d = .54$). This may be because head movement affects focusing attention and the meaning of yes/no answer. Anthropomorphism of a robot is not simply about its appearance, but about how participants perceive it as a human-like interlocutor. This perception can enhance participants' social experiences when interacting with robots. People naturally interact with robots in ways similar to how they engage with human partners from everyday interactions. As social partners, the lifelike qualities of robots, enhanced by head movements, may help them become more natural in HRI.

In terms of likeability (H1c) and perceived intelligence (H1d), all types of head movements were found to have a significant effect on likeability ($t(89) = 21.92$, $p < .001$, $d = .90$) and perceived intelligence ($t(89) = 21.78$, $p < .001$, $d = .88$) compared to non-movement. These results reveal that head movements can create positive and smart-looking impressions of robots. The three types of movements used in this experiment were adapted from typical human gestures. However, there is a wide variety of head movements that people use in communication. The match between the type of movement and the context of the conversation may influence the creation of positive and smart-looking impressions of the robot. This suggests that selecting the appropriate type of movement for the conversational context is important for creating positive and smart-looking impressions of robots.

Second, the timing of the head movement prior to utterance had a significant effect on perceived naturalness in human-robot conversations (H2). We observed a positive perceived naturalness rating with the robot's head movement prior to utterance ($t(69.72) = 4.55$, $p < .001$, $d = .58$) compared to head movement simultaneous with utterance. This suggests that head movement prior to utterances can reduce the long response delay problem between utterances. However, in our approach to head movement timing, it is technically difficult to eliminate all response delays before the robot's movement. This is because the robot requires some time to recognize and understand natural language in human speech in order to select the appropriate type of head movement for the conversational context. While the timing of head movements cannot be shortened to occur before the robot's understanding stage, the response delay during the robot's speech generation stage can be reduced.

Additionally, perceived naturalness was significantly correlated with likeability (H2-1a) and perceived intelligence (H2-1b). Likeability of robots is affected by other variables at the same time [88]. These results support the notion that perceived naturalness is one of the main variables determining the likeability of robots and is related to smart-looking impressions. This approach, which coordinates the separate timing of head movements and robot utterances, is fundamentally different from much of the research that has focused on reducing the response delay of robots performing feedback on human intentions as quickly as possible.

5.1 Theoretical Implications

Several studies have revealed the effects of nonverbal expressions in HRI-this includes not only head movement but also various other expressions (e.g., posture, arm gesture, and facial expression), through the investigation of human-like movements in humanoid robots. However, there are limitations to the behavioral expressions of non-humanoid robots. A few studies have investigated the effects of minimized nonverbal expressions, for example, eye gazing in non-humanoid robots [14] and showed the potential of minimal gestures in the interaction of humans and non-humanoid robots.

In this study, we focused on the head movement of a robot as a minimal nonverbal expression that can be easily implemented in not only non-humanoid but also humanoid robots. Our findings demonstrate the effects of minimal head movements on user perception in HRI. It may be useful to develop a robot interface that enhances the social abilities of robots.

In addition, our study revealed that head movements prior to utterances provided a higher perceived naturalness rating than head movements simultaneous with utterances. This proves the importance of the timing of head movement and provides a direction on how to enhance HRI through timing adjustment of movement.

5.2 Practical Implications

To summarize, nonverbal expressions should be considered for enhancing the social abilities of robots. Human gestures are typically multimodal behaviors. However, we should consider the limitations of the embodiment in non-humanoid robots. This study showed that minimal head movements of a simple-shaped robot significantly improved HRI. Technically having more human-like movements in humanoid robots is not a fundamental solution for making HRI smoother. In contrast, even minimal and simple movements can make the interactions between humans and robots more natural. Our study presents an example of a newly designed robot with a simple-shaped body and minimal head movement mechanism. The simplicity of the robot's design makes it easily implementable for other robots.

Robots with minimal movements and simple shapes can be easily integrated into daily life. Compared with the human-like multimodal gestures of humanoid robots, they incur less burden and can easily blend into various environments.

Our findings also indicate that timing adjustment can benefit natural human-robot conversations. We expected that the timing of head movement prior to utterance could reduce the problem of long response delays between speeches in human-robot conversations. People feel unnatural while waiting for a robot's response after talking to it. The results revealed that the robot's head movement prior to the utterance could reduce the long response delay problems. Therefore, the timing of the head movements is important in HRI and can make it more natural. It is necessary to consider head movements and their timing when designing and studying the physical interactions of social robots.

5.3 Limitations and Future Research

In the experiment of this study, there was a lack of engagement in the conversation, because the participants had only brief interactions with the robot according to the script. During conversations in human interactions, people can fulfill others' immediate intentions and purposes [74]. A certain level of engagement is required to create interpersonal communication [35], which is closely related to relationships and social contexts [75]. In the previous study [76], engagement of HRI was measured on the basis of participants' emotional involvement and feeling of strong interaction with robots. Future studies should include the social engagement of HRI in their experiments to investigate the social abilities of robots more clearly. Additionally, the approach of this study is applicable to other types of robots, and its results can be compared with those of other robots.

Moreover, the experiment in this study was conducted in a virtual simulation environment. To investigate the effects of the head movement of the robot, video recordings of the robot's responses were prepared in advance. However, the participants had an onscreen conversation with the robot; therefore, their concentration might not be on the same level as in a real human-robot conversation. Future studies should be conducted in more realistic environments.

Additionally, in this study, we manipulated three types of head movements: nodding (A), shaking (B), and tilting (C), which are widely used and typically understood in human interaction. However, there are many other types of head movements in human behavior that were not explored here. Future studies should investigate additional robot head movements based on a broader range of human head movements and examine their impact on user perception in HRI. Moreover, cultural differences can be seen in head gestures [3, 77]. The cultural diversity of gestures in HRI should also be considered. Therefore, it is necessary to identify representative cross-cultural gestures in HRI or to localize the robot gestures according to cultural differences.

6 Conclusion

In the experiment, we hypothesized that head movement positively affects the participants' perceptions of the robot. The experimental results supported our hypothesis. To investigate different types of head movements, we manipulated three types of head movements: nodding (A), shaking (B), and tilting (C). In all types of head movements, when the robot performed head movement, the participants' perceptions of anthropomorphism, animacy, likeability, and intelligence were higher than in the non-movement (utterance only) condition.

With regard to head movement timing, we expected that head movement prior to utterance would show a higher perceived naturalness rating than head movement simultaneous with utterance. The experimental results supported our hypothesis. There was a positive increase in the perceived naturalness rating for head movements prior to the utterance condition. We also hypothesized that perceived naturalness would positively correlate with likeability and perceived intelligence. The results revealed that perceived naturalness significantly correlated with likeability and perceived intelligence.

In summary, the head movements of the robot increased the participants' perceptions of anthropomorphism, animacy, likeability, and perceived intelligence of the robot compared with the non-movement condition. Second, the timing of head movement prior to utterance had a positive effect on perceived naturalness compared to head movement simultaneous with utterance. Third, perceived naturalness was positively correlated with the likeability and perceived intelligence of the robot. The results of this study suggest that a robot's head movements, such as nodding, shaking, and tilting, play significant roles in HRI, and adjusting the timing of these movements is a crucial aspect of designing and studying HRI.

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